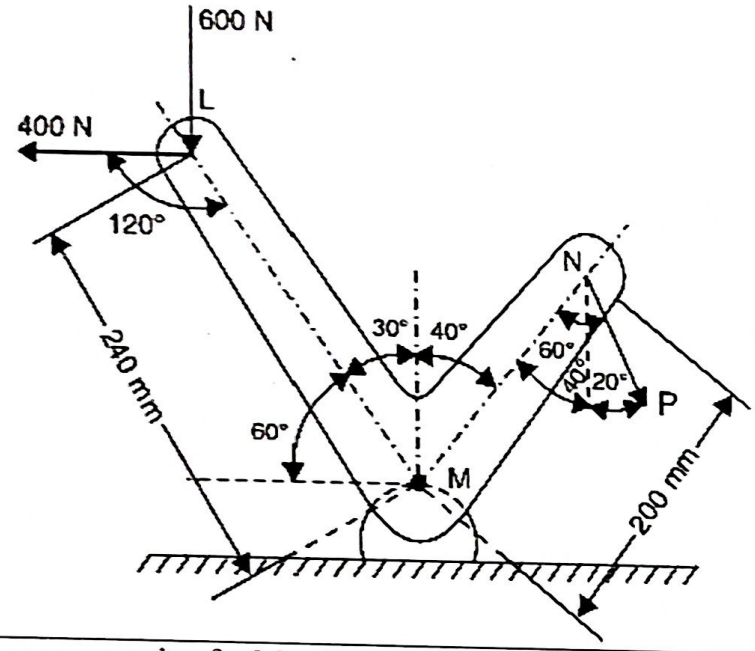
Sumit kumar
24G1031

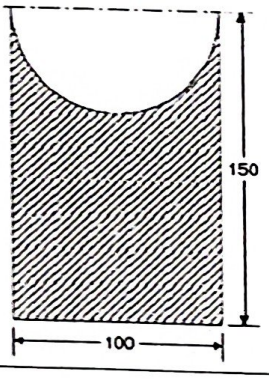
MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF CIVIL ENGINEERING
End Term Examination, May, 2025.

Course: B.Tech. Semester: II Branch: All Branches.
Subject: Engineering Mechanics Subject Code: CE 24103
Time: 2 hours. Maximum Marks: 40

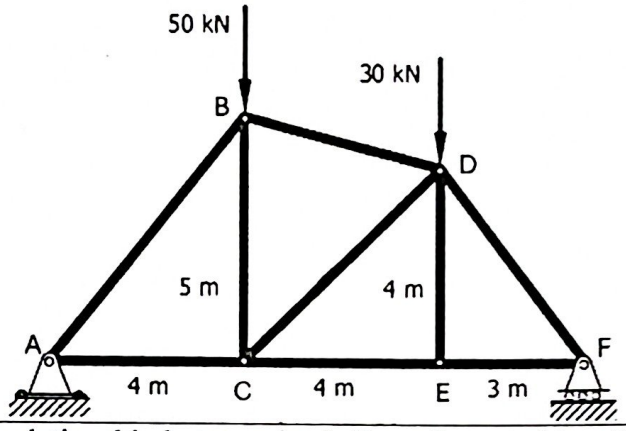
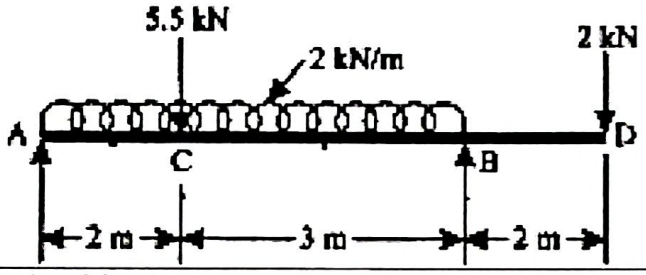
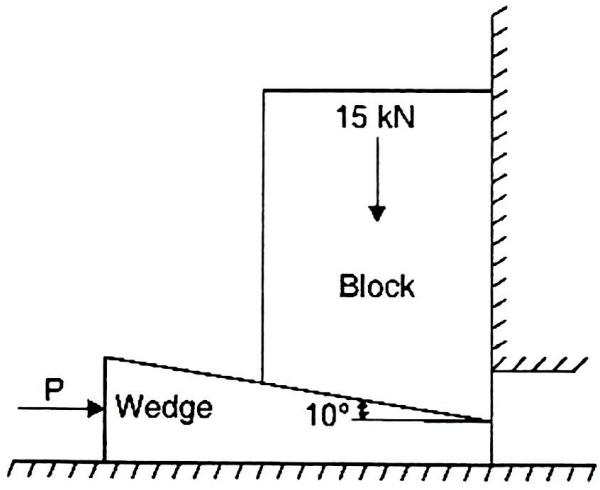
Note: - Answer all questions. Make suitable assumptions if necessary.

Q.No.	Question	Marks	CO/BT
Q.1.	a) Write short note on "The principle of transmissibility of forces" with suitable sketches.	2	CO 1, CO 2
	b) The lever LMN of a component of a machine is hinged at M, and is subjected to a system of coplanar forces as shown in figure. Neglecting friction determines: (i) The magnitude of the force P to keep the lever in equilibrium. (ii) The magnitude and direction of the reaction at M.	6	BT 1,2,3
Q.2.	a) Derive an expression for Moment of Inertia of a Circle about its diameter.	2	CO3
	b) A semicircular cut is being provided from a rectangular beam as shown in figure. Determine the moment of inertia of the shaded lamina about Horizontal Centroidal Axis. All dimensions are in mm.	6	BT 3,4





P.T.O.....

Q.3	a) Differentiate between method of joints and method of sections of trusses with suitable sketches. b) A truss is loaded as shown in figure. Find out the magnitude & nature of forces in all the members of the truss and tabulate the results. 	2 6	CO 4 BT 4.5
Q.4	a) Derive the relationship between load, Shear force and bending moment. b) Calculate the support reactions and draw SFD and BMD for the beam loaded as shown in figure. 	2 6	CO 5 BT 5,6
Q.5	a) Define Angle of friction and Angle of repose with suitable figure. b) A block overlying a 10° wedge on a horizontal floor and leaning against a vertical wall and weighing 15 kN is to be raised by applying a horizontal force to the wedge. Assuming the co-efficient of friction between all the surfaces in contact to be 0.3, determine the minimum horizontal force, to be applied to raise the block 	2 6	CO 6 BT 5